ICSE 2027 – Biology (Science Paper 3)

Excretory System – Past Paper Question Bank

Rules: attempt every question cold, on paper, with no notes. Then mark against the scheme, write your score in the last column, and log every gap. Re-attempt the questions you missed after 2–3 days.

A. Definitions and one-liners

#	Question	Paper	Marks	Score
1	Define ultrafiltration.	2024 Q8(i)	1	
2	Explain the term Excretion.	2025 Q8(i)	1	
3	What is Excretion?	2026 Spec Q4(i)	1	
4	What is Micturition? Explain.	2025 Q5(i)	1	
5	Name the main nitrogenous waste formed in the body.	2024 Q2(i)(d)	1	
6	Name the duct which carries urine from the urinary bladder to outside the body.	2024 Spec Q2(i)(d)	1	
7	Name the pigment responsible for the pale yellow colour of normal human urine. (MCQ)	2023 Q1(ix)	1	
8	A person with kidney failure has proteins in the urine. What is this condition called? (MCQ)	2026 Q1(xii)	1	

B. Structure, location and comparison

#	Question	Paper	Marks	Score
1	Renal cortex has a dotted appearance and renal medulla has a striped appearance. Explain.	2023 Q7(ii)	2	
2	Which parts of the nephrons are present in the renal cortex and the renal medulla? (figure MCQ)	2027 Spec Q1(viii)	1	
3	Mention the regions of the kidney where we find the glomerulus and Henle's loop.	2026 Spec Q7(iv)	2	
4	Study the diagram (renal artery, renal vein, pelvis, ureter, cortex, medulla) and match structure with function.	2026 Q2(v)	5	
5	Arrange in logical sequence: Loop of Henle, Distal convoluted tubule, Bowman's capsule, Proximal convoluted tubule.	2024 Q2(ii)(d)	1	
6	Match: Kidney with Nephrons (Column I with Column II).	2024 Spec Q2(v)	1	
7	Differentiate between afferent arteriole and efferent arteriole (diameter).	2024 Q3(iii)	2	
8	State the difference between nephron and neuron.	2025 Q8(ii)	2	
9	Differentiate between the functions of urethra and uterus.	2025 Q5(iii)	2	

C. Diagram

#	Question	Paper	Marks	Score
1	Draw a neat, labelled diagram of a Malpighian capsule.	2023 Q6(v)	3	

D. Odd one out (1 of 5 marks each)

#	Question	Paper	Marks	Score
1	Odd one out + category: Uterus, Urethra, Urinary bladder, Ureter.	2023 Q2(iv)(c)	1	
2	Odd one out + category: Tonsils, Glomerulus, Spleen, Lymph nodes.	2024 Q2(iv)(b)	1	
3	Odd one out + category: Renal pelvis, Renal artery, Renal cortex, Renal medulla.	2024 Spec Q2(iv)(b)	1	
4	Odd one out + category: Urochrome, Urea, Keratin, Uric acid.	2024 Spec Q2(iv)(c)	1	
5	Odd one out + category: Urochrome, Urea, Uric acid, Nucleic acid.	2026 Q2(iii)(c)	1	
6	Odd one out + category: Ureter, Urethra, Uterus, Renal pelvis.	2026 Spec Q2(iii)(d)	1	

E. Application and case-based

#	Question	Paper	Marks	Score
1	Level of urea in the renal artery and renal vein after a protein meal. (MCQ)	2027 Spec Q1(iv)	1	
2	Urine output of a patient suffering from severe cholera. (MCQ)	2027 Spec Q1(v)	1	
3	Assertion-Reason: frequent urination in summer than in winter.	2024 Spec Q1(vi)	1	
4	Prostate gland case: (a) location (b) why alkaline secretion (c) structure regulating urine flow from bladder into urethra.	2026 Q4(v)	3	

F. ADH link

#	Question	Paper	Marks	Score
1	Hormone that affects urination. (MCQ)	2024 Q1(iv)	1	
2	Assertion-Reason: deficiency of ADH causes increase in urination.	2025 Q1(ix)	1	
3	Differentiate between Diabetes mellitus and Diabetes insipidus with reference to the hormone involved.	2026 Spec Q4(ii)	2	

What the pattern tells you

- Odd-one-out and the cortex vs medulla distinction appear almost every year. Be able to sort urinary parts, renal parts, nitrogenous wastes and urine components instantly.

- The Malpighian capsule diagram has already carried 3 marks and the syllabus stresses it. Draw it until it is automatic.
- Selective reabsorption, tubular secretion and urine vs plasma composition are in the syllabus but barely appear in these papers, apart from ultrafiltration once. That makes them the likeliest gap an examiner could exploit.

Gap log

Questions I missed / concepts to fix: